



IIT DELHI
MASTER IN TECHNOLOGY, ROBOTICS

END-TERM MTECH PROJECT PART-II EVALUATION

Residual Human–Robot Interaction Modeling

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GUIDE :- PROF. SHUBHENDU BHASIN AND PROF. SITIKANTHA ROY

DATE :- 21st May 2026

Problem statement

Residual Human–Robot Interaction Modeling

- The neural network to learn the part of Human–Robot Interaction that OpenSim did NOT model but happened during experiment

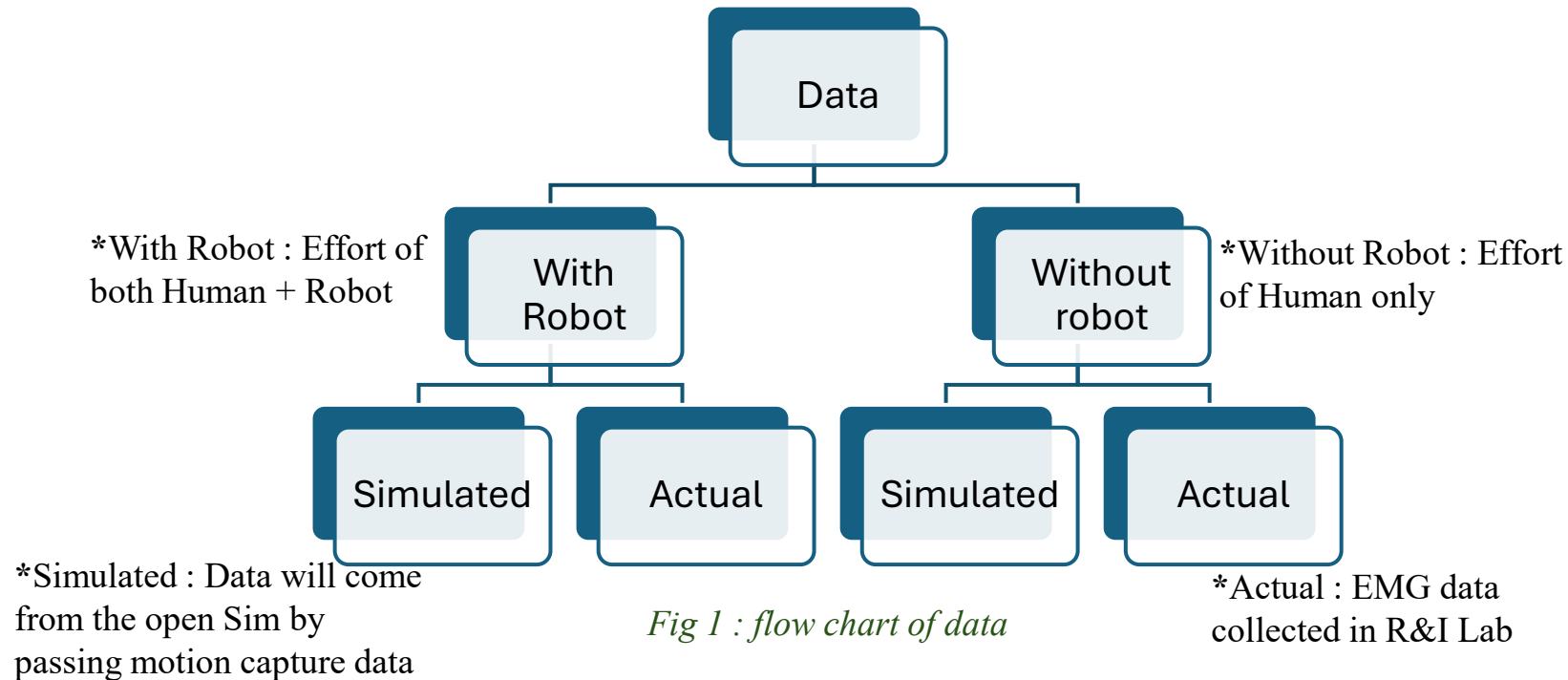


Fig 2 : person doing experiment

Methodology

Collected Data Upper Limb

1. **Motion Capture:**
Vicon Nexus tracks body movements.
2. **Activity:** Biceps Curl
3. **DOF:** 1.
4. **Muscle Involve:** 6
5. **Exosuit** by JATC Lab



Vid 1: Data Collection video

ANGLE

info

- record_name
- type
- created_with_version
- exported_with_version
- last_name
- first name
- sex
- measurement_date

movement

- sources
 - signals
 - signal_1,
 - *data*, <!-- Elbow -->
 - signal_2
 - *data*, <!-- Shoulder -->
 - units
 - name
 - type
 - name
 - time_begin
 - time_end

EMG

info

- record_name
- type
- created_with_version
- exported_with_version
- last_name
- first name
- sex
- measurement_date

movements

- sources
 - name
 - signals
 - signal_1
 - *data* <!--Data for emg1 -->
 - signal_2
 - *data* <!--Data for emg2 -->
 - signal_3
 - *data* <!--Data for emg3 -->
 - signal_4
 - *data* <!--Data for emg4 -->
 - signal_5
 - *data* <!--Data for emg5 -->
 - signal_6
 - *data* <!--Data for emg6 -->
 - type
 - name
 - time_begin
 - time_end

Methodology

Data Pipeline

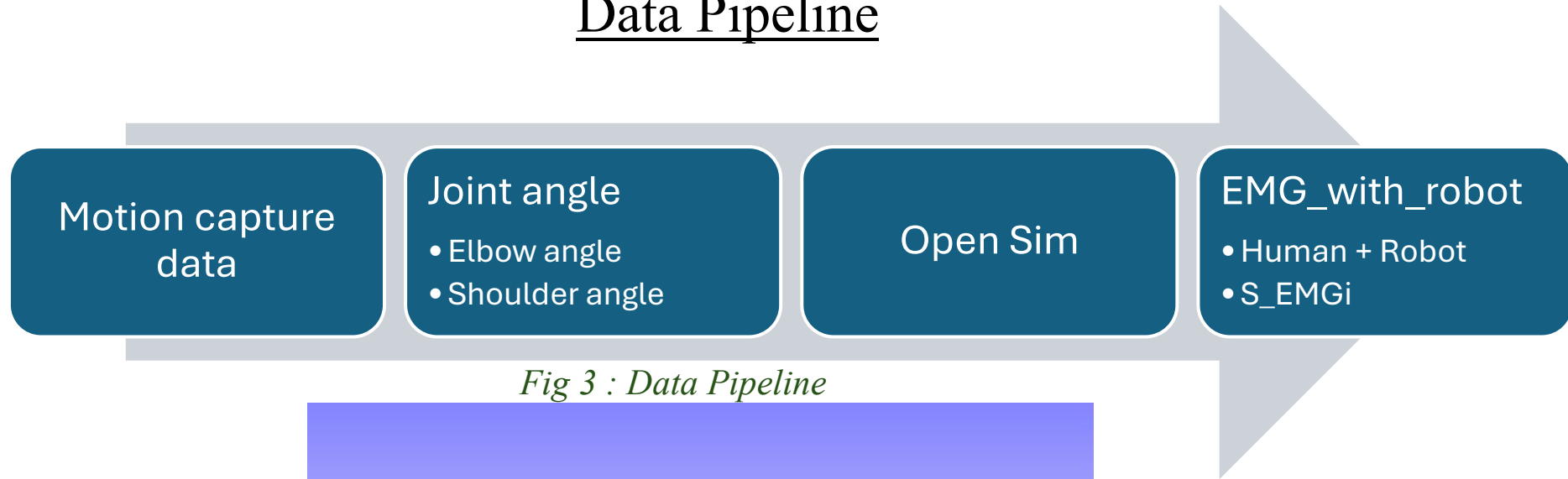
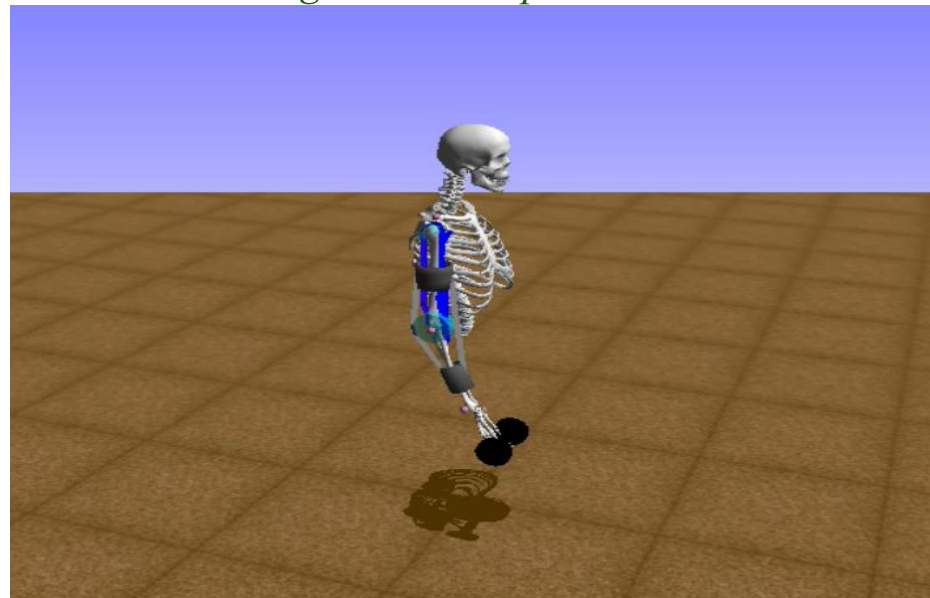


Fig 3 : Data Pipeline



Vid 2 : Simulation video

Methodology

EMG Signal Processing

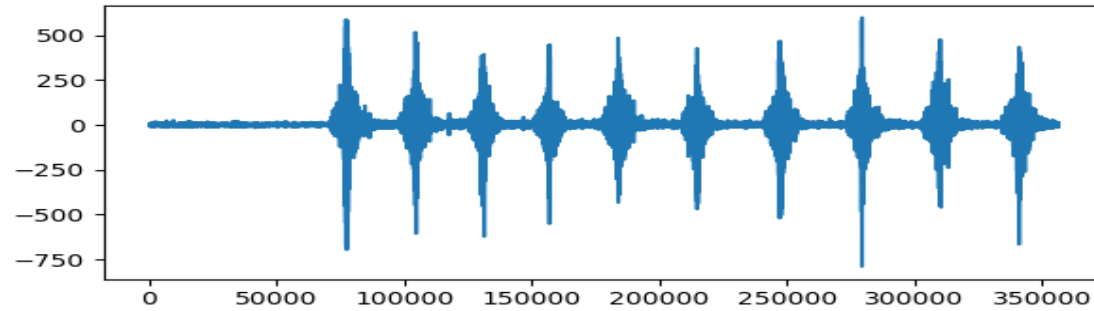


Fig 5 : Raw EMG

Notch Filter

Band Pass
Filter

Applying MViC

Down
sampling

Fig 7 : Signal processing pipeline

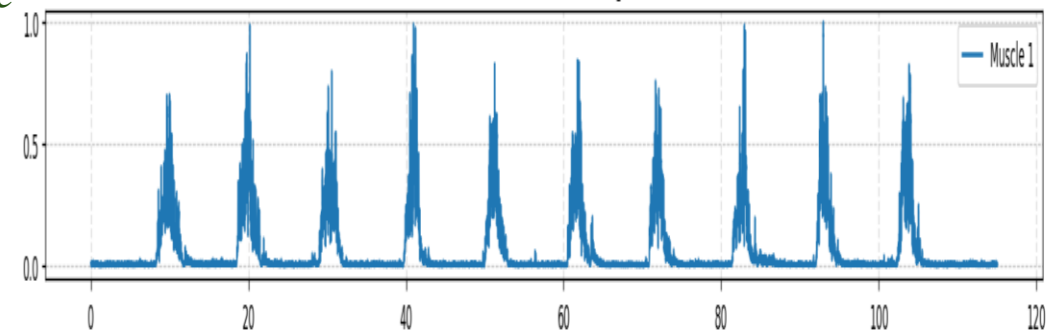


Fig 6 : Processed EMG

Methodology

Model Input : Output

Model Inputs:

[θ_e : Elbow angle
 θ_s : Shoulder angle
 $\dot{\theta}_e$: Elbow velocity
 $\dot{\theta}_s$: Shoulder velocity
 s_emg1 : Simulated EMG of biceps long
 s_emg2 : Simulated EMG of biceps short
 s_emg3 : Simulated EMG of triceps long
 s_emg4 : Simulated EMG of triceps lateral
 s_emg5 : Simulated EMG of triceps medial
 s_emg6 : Simulated EMG of brachialis
weight : external weight
]

Model Output:

The model predicts the Human–Robot Interaction (HRI) residual, defined as:

$$\Delta = (a_emg_i - s_emg_i)$$

This is predicted separately for each muscle:

- Δ_emg1 : biceps_long
- Δ_emg2 : biceps_short
- Δ_emg3 : triceps_long
- Δ_emg4 : triceps_lateral

Methodology

Model Architecture

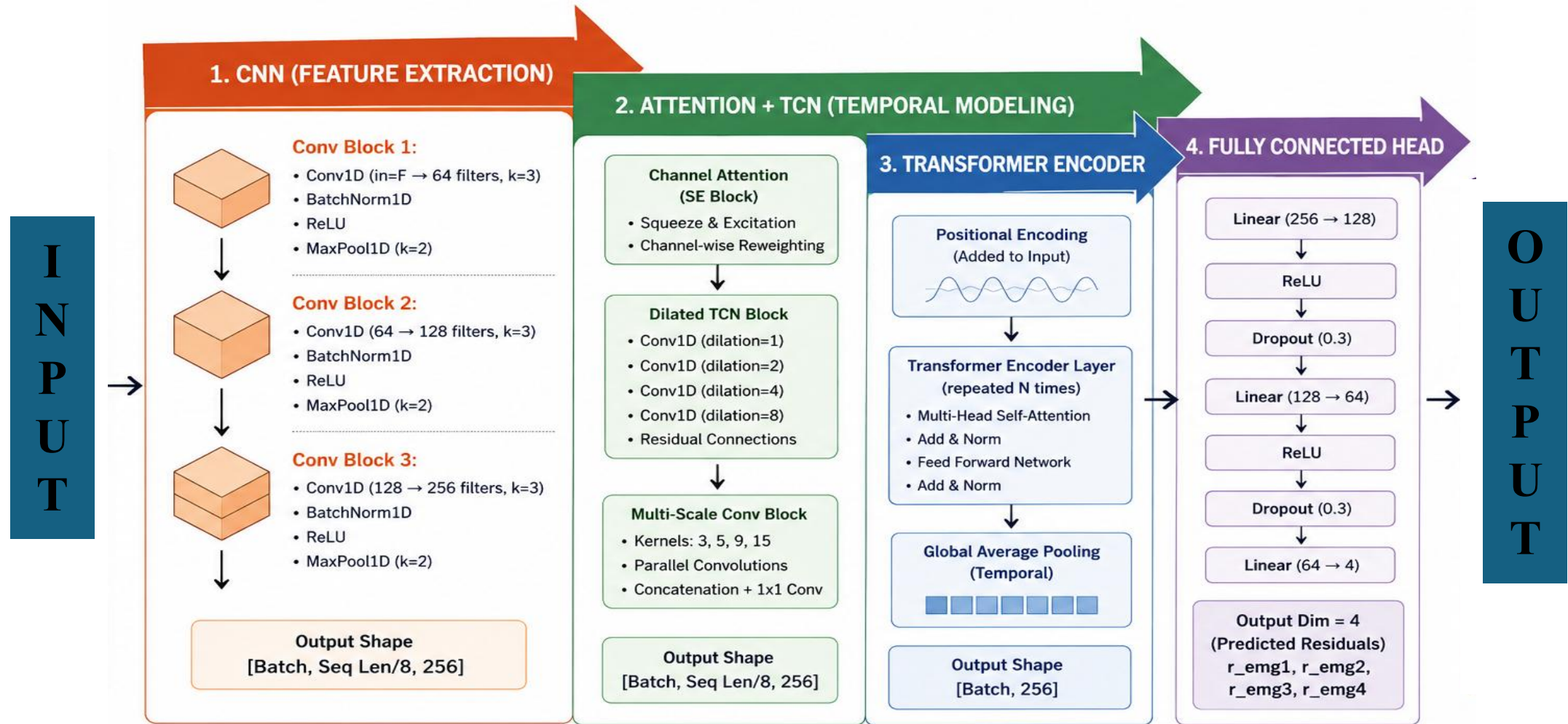


Fig. 8 : Model Architecture

Results

Train and Validation Results

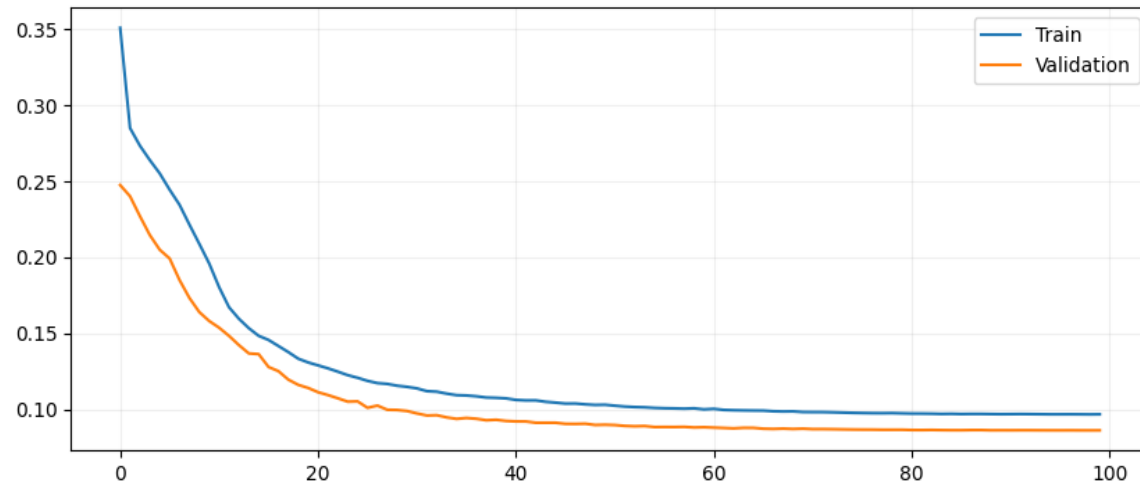


Fig. 9 : Training Curve



Fig. 10 : R2 score compression

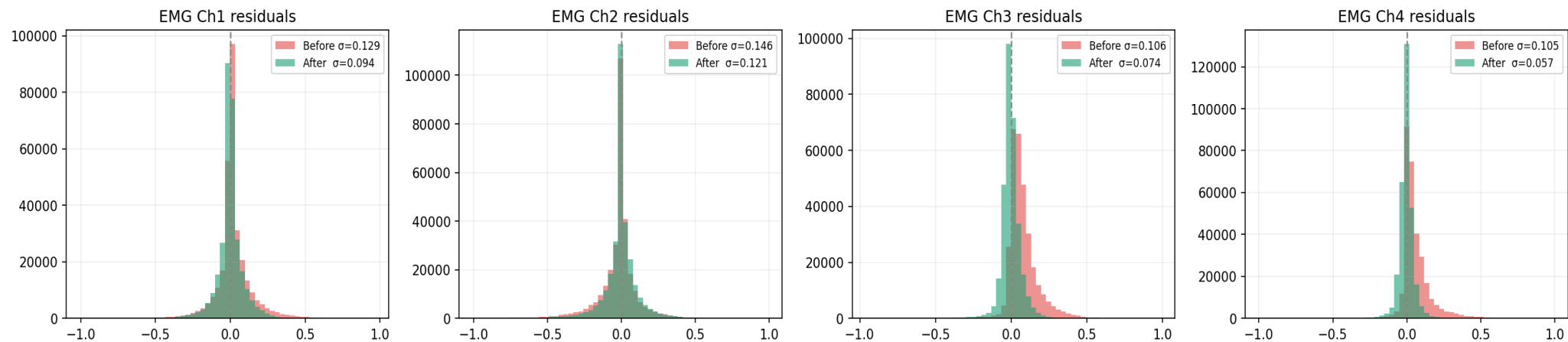
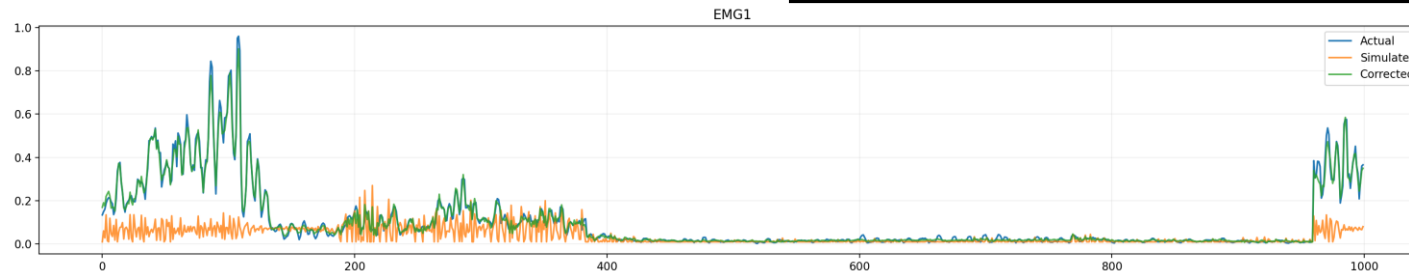


Fig. 11 : Residual Distributions

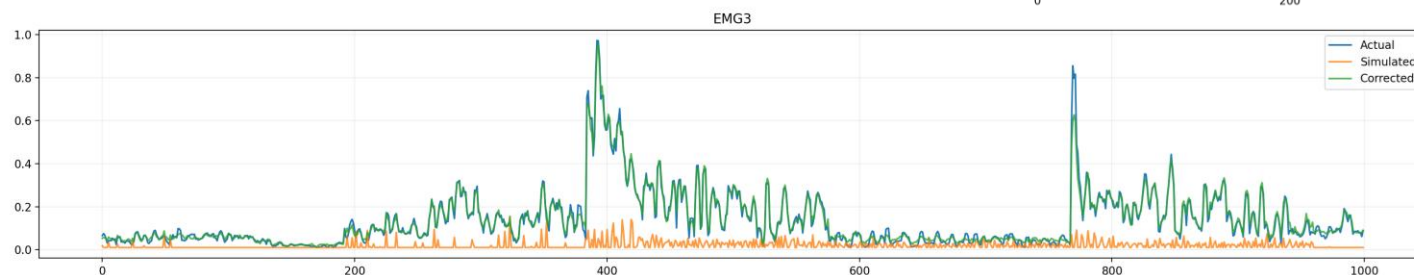
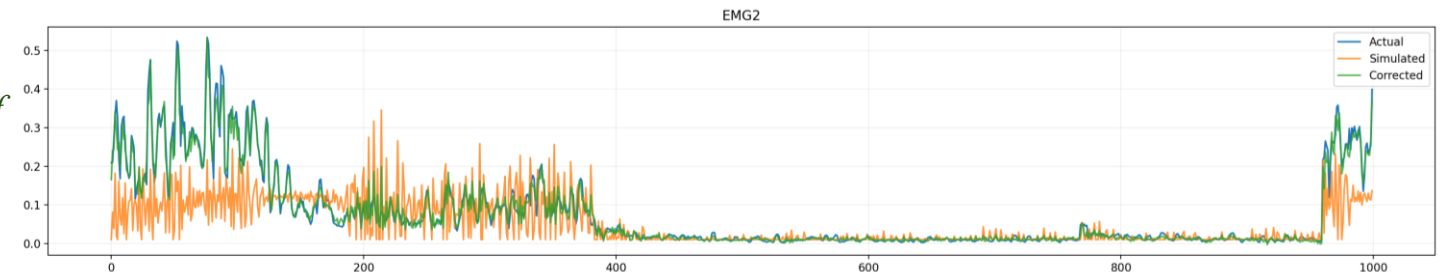
Results

Train and Validation Results



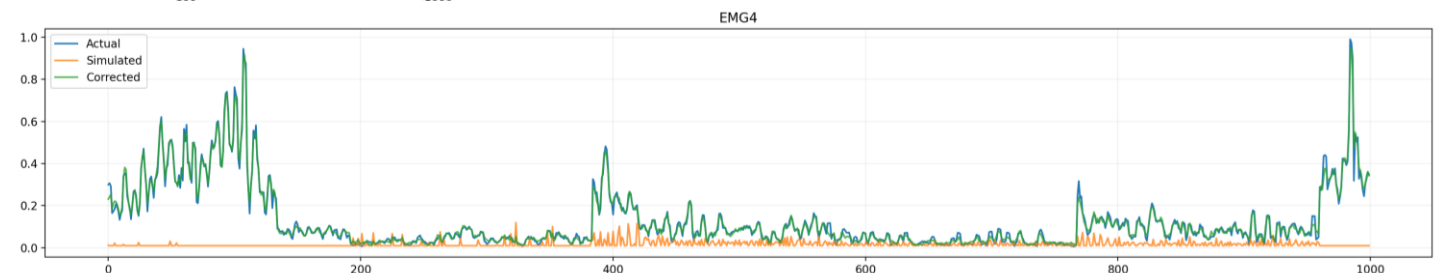
*Fig. 12 : Temporal Comparison of Actual, Simulated, and Corrected **EMG1** Signals*

*Fig. 13 : Temporal Comparison of Actual, Simulated, and Corrected **EMG2** Signals*



*Fig. 14 : Temporal Comparison of Actual, Simulated, and Corrected **EMG3** Signals*

*Fig. 14 : Temporal Comparison of Actual, Simulated, and Corrected **EMG4** Signals*



Results

Train and Validation Results

After correction:

- **EMG4** is the star: $R^2=0.969$, $r=0.984$. The model has almost perfectly recovered the actual signal.
- **EMG3**: $R^2=0.936$, $r=0.968$. Excellent.
- **EMG1**: $R^2=0.867$, $r=0.935$. Very good.
- **EMG2**: $R^2=0.682$, $r=0.858$. Good, but still the hardest channel.

OVERALL RESULTS :

RMSE: 0.03839

MAE : 0.01685

R2 : 0.86347

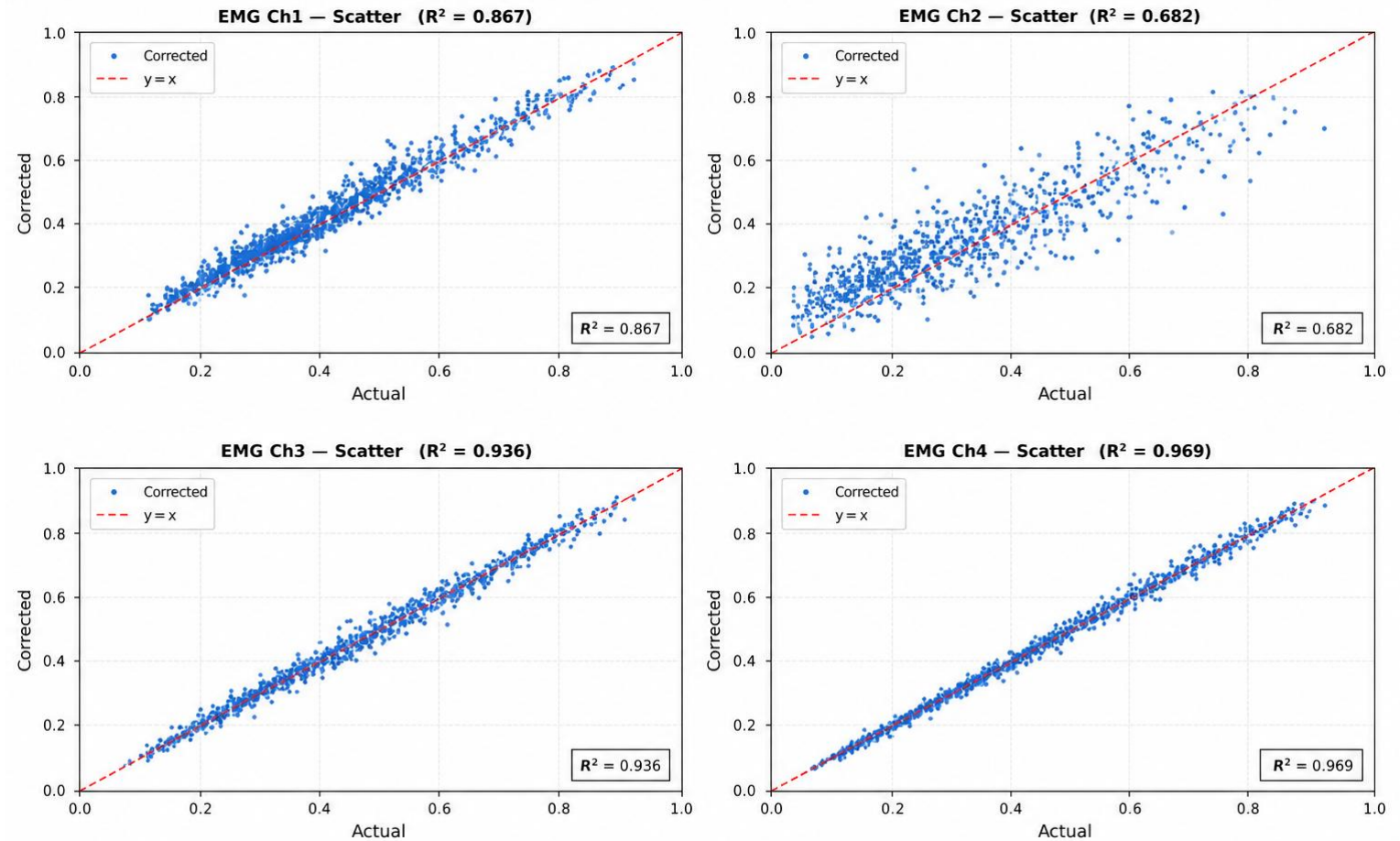
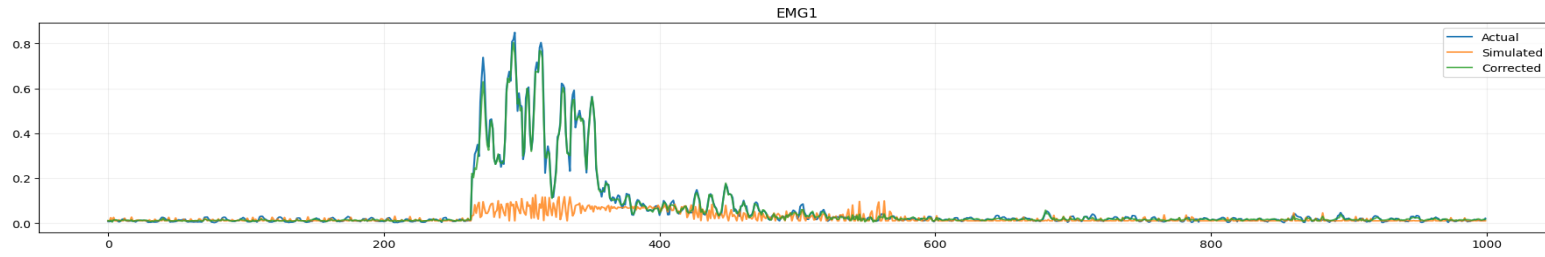


Fig. 15 : Actual vs Corrected EMG Signals

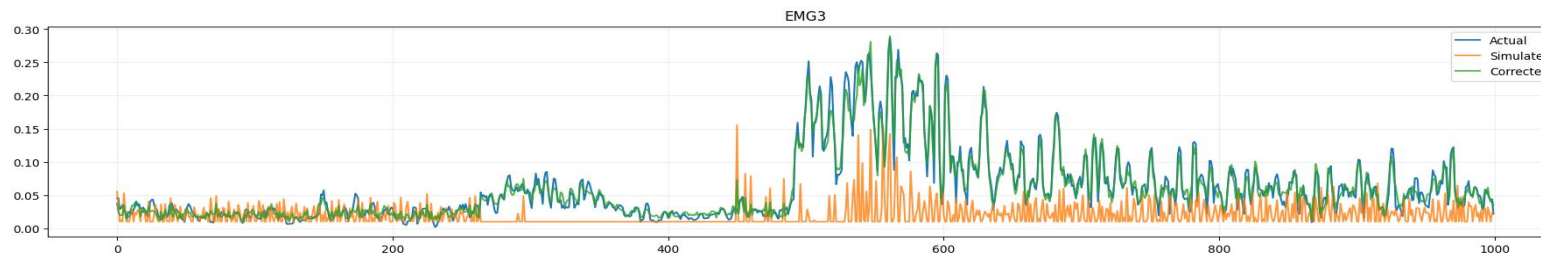
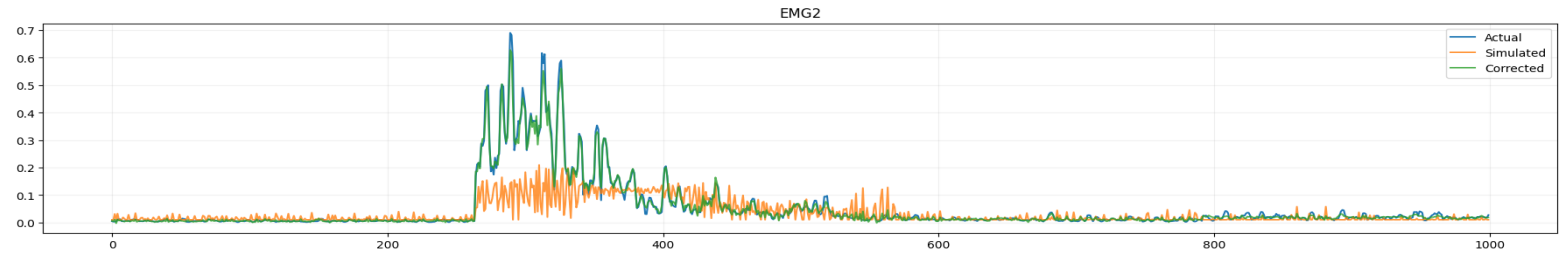
Results

Test Results



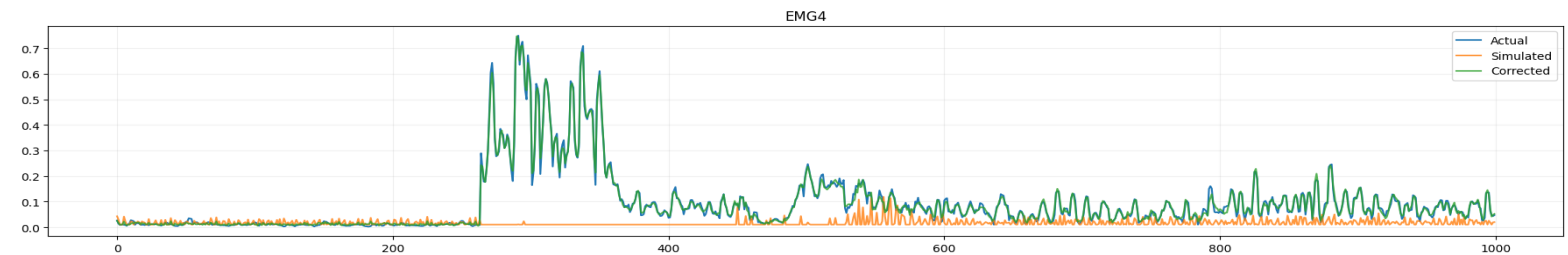
*Fig. 16 : Temporal Comparison of Actual, Simulated, and Corrected **EMG1** Signals*

*Fig. 17 : Temporal Comparison of Actual, Simulated, and Corrected **EMG2** Signals*



*Fig. 18 : Temporal Comparison of Actual, Simulated, and Corrected **EMG3** Signals*

*Fig. 19 : Temporal Comparison of Actual, Simulated, and Corrected **EMG4** Signals*



Thank You ...